

PART 1 - Does Standalone Solar Still Make Sense in the New ToD Era?

For the last several years, the C&I renewable-energy conversation in Maharashtra was relatively straightforward:

Install Solar → Replace Grid Power → Reduce Energy Cost

That equation is becoming more nuanced.

With the evolving Time-of-Day (ToD) tariff framework, the value of electricity increasingly depends not only on how much power a consumer uses, but also on when it is consumed. Solar remains one of the lowest-cost renewable sources, but its generation is concentrated during daylight hours.

This changes the question from:

“Is solar cheaper than grid power?”

to:

“Does the solar generation profile replace the hours in which grid power has the highest value?”

For industrial consumers with strong daytime demand, standalone Solar can continue to deliver compelling economics. However, where a meaningful share of consumption occurs in non-solar or higher-value ToD periods, Solar alone can leave a large part of grid consumption untouched.

BESS can bridge this mismatch by storing renewable energy and shifting it to periods where it creates greater value. But storage also introduces battery cost, round-trip losses, degradation and financing cost. The commercial question is therefore not whether BESS works technically; it is whether the value created by shifting energy exceeds the cost of storing it.

Illustrative comparison: Solar vs Solar + 4-hour BESS

Parameter	Solar CAPEX	Solar Group Captive	Solar + BESS CAPEX	Solar + BESS Group Captive
Customer consumption	(Uniform on RTC basis)			
Configuration	4.7 MWp Solar	4.7 MWp Solar	8 MWp Solar + 11.5 MWh BESS	8 MWp Solar + 11.5 MWh BESS
RE / Grid replacement	35.1%	35.1%	56.4%	56.4%
Total landed RE cost	₹4.67/kWh	₹5.68/kWh	₹6.29/kWh	₹7.26/kWh
Balance grid consumption	64.9%	64.9%	43.6%	43.6%

Weighted average saving (on total consumption)	₹1.45/kWh	₹1.10/kWh	₹1.80/kWh	₹1.25/kWh
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Illustrative scenario - key assumptions

- Location: Maharashtra; analysis based on an indicative industrial consumption profile.
- Business models compared: CAPEX and Group Captive.
- Standalone Solar configuration: 4.7 MWp. Solar + BESS configuration: 8 MWp Solar + 11.5 MWh BESS.
- BESS cost assumption: approximately ₹5.50/kWh of energy delivered through storage.
- Transmission charge assumption: ₹1.02/kWh.
- Applicable losses, Electricity Duty, ToD tariffs and banking treatment are as used in the underlying Maharashtra model.
- Indicative lapse / unutilised renewable energy: approximately 4.5% for Solar and 4.6% for Solar + BESS.
- All values are illustrative; actual outcomes vary with customer load profile, site conditions, tariff, financing, contract terms and regulatory treatment.

Solar + BESS improves renewable replacement from 35.1% to 56.4% in the illustrative case - but the battery must earn back its additional cost through better ToD utilisation.

The commercial answer therefore depends primarily on three variables:

- Customer load profile and the share of consumption falling in solar / peak / non-solar hours;
- The effective ToD differential and avoided grid / DG cost; and
- Battery cost, cycling, round-trip efficiency and expected utilisation.

Our view is simple: do not start by asking how much battery to install. Start by identifying which hours of grid consumption are actually worth replacing.

In Part 2, we look at why adding Wind can materially change this equation - and why optimising the Wind-Solar mix can sometimes reduce the storage requirement itself.